

// Day 11 of 30

STORY + LESSON

How a Single SQL Query Crashed Our Production Server

→ Swipe through all 6 slides



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The Innocent Query

It looked harmless. It wasn't.

```
SELECT * FROM transactions
WHERE created_at > '2025-01-01'
ORDER BY amount DESC;

-- Looked fine in dev (100 rows)
-- Production had 8 MILLION rows
```

💡 Always test queries with production-sized data



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What Happened Next

3 AM. PagerDuty goes off.

CPU: 98%

Memory: 95%

Response time: 47 seconds

Active connections: maxed out

- One query locked the entire table
- Every other query queued behind it

💡 One bad query can take down everything



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Why It Crashed

No index. Full table scan.

EXPLAIN ANALYZE showed:

```
-- Seq Scan on transactions
-- Rows examined: 8,247,312
-- Time: 43,291ms
```

No index on created_at column

No LIMIT clause

SELECT * instead of specific columns

💡 EXPLAIN ANALYZE is your best friend



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The 5-Minute Fix

Three changes. Problem solved.

1. Added index:

```
CREATE INDEX idx_txn_date  
ON transactions(created_at);
```

2. Added LIMIT:

```
LIMIT 1000
```

3. Selected specific columns:

```
SELECT id, amount, status
```

💡 Index + LIMIT + specific columns = safe



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Before vs After

Numbers don't lie

BEFORE:

Query time: 43,291ms
Rows scanned: 8,247,312
CPU: 98%

AFTER:

Query time: 12ms
Rows scanned: 1,000
CPU: 3%

💡 3,600x faster with 3 simple changes



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Lessons Learned

Never again.

1. Never use `SELECT *` in production
2. Always add indexes on `WHERE` columns
3. Always use `LIMIT` on large tables
4. Run `EXPLAIN` before deploying queries
5. Test with production-sized datasets
6. Set up query timeout limits

💡 Prevention > firefighting at 3 AM



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done = true

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